LAB EXPERIMENT RECORD

Set 16: Customer Demographics Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424049
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Day	Saturday

AIM

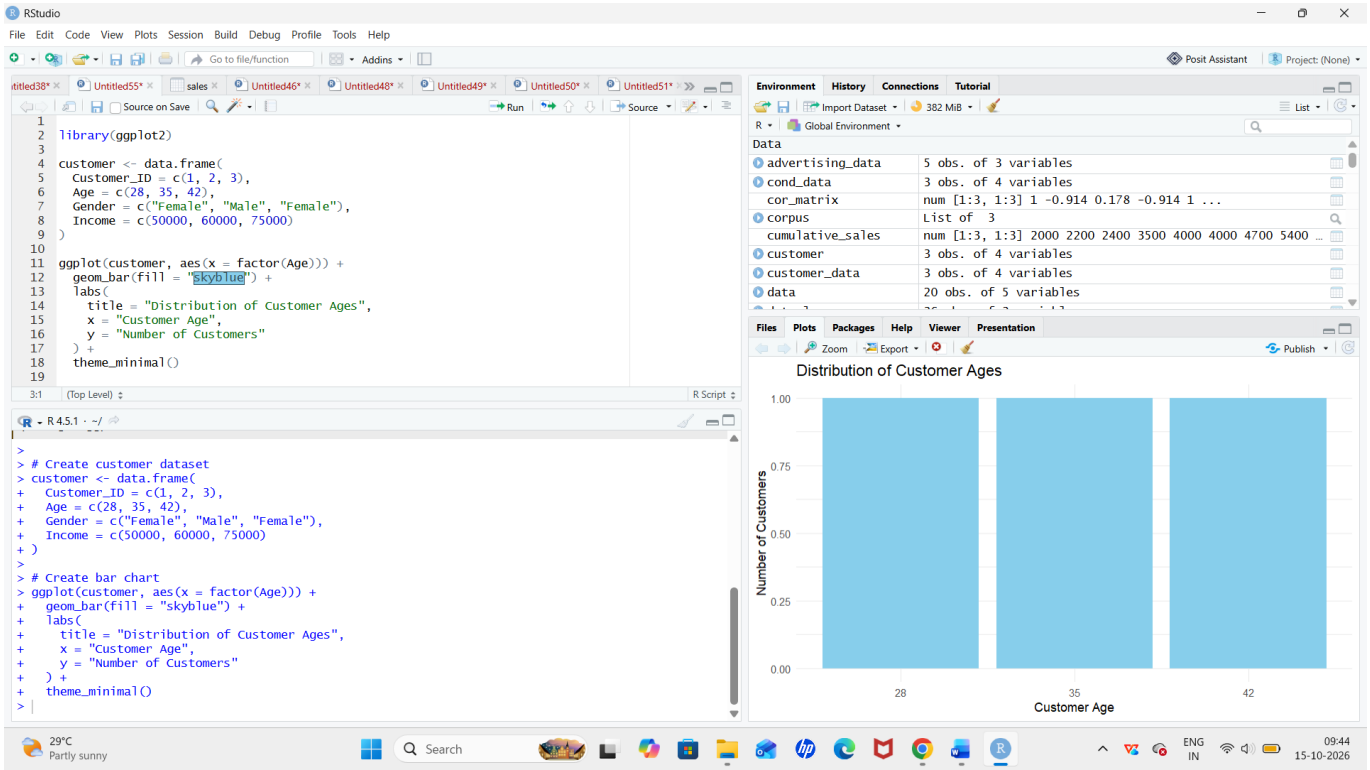
To analyze and visualize customer demographic data using R programming by creating a bar chart to display the distribution of customer ages, a pie chart to represent gender distribution, and a table to display customer demographic information. To combine these visualizations into an interactive Tableau dashboard for effective demographic analysis..

1. Create a bar chart to visualize the distribution of customer ages.

```
library(ggplot2)

customer <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

ggplot(customer, aes(x = factor(Age))) +
  geom_bar(fill = "skyblue") +
  labs(
    title = "Distribution of Customer Ages",
    x = "Customer Age",
    y = "Number of Customers"
  ) +
  theme_minimal()
```



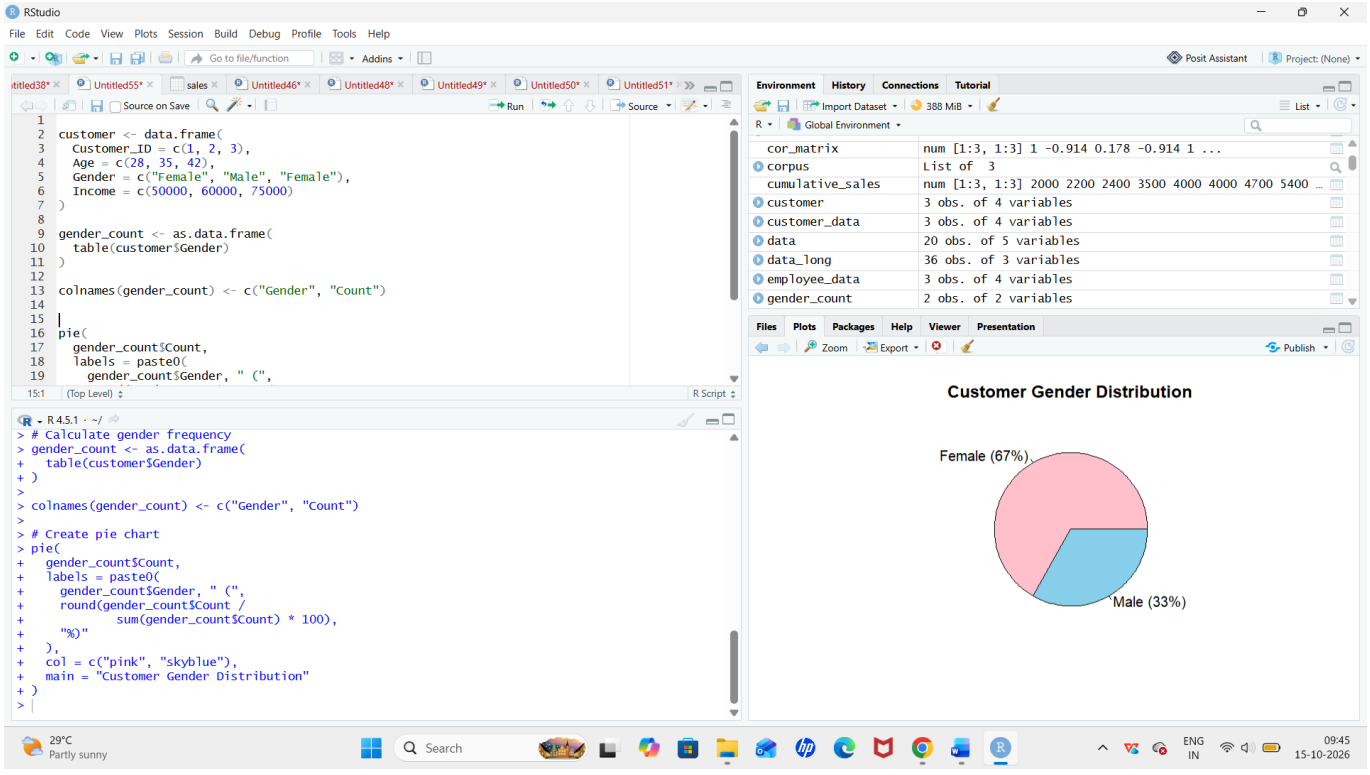
2. Generate a pie chart to represent the distribution of customers by gender.

```
customer <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

gender_count <- as.data.frame(
  table(customer$Gender)
)

colnames(gender_count) <- c("Gender", "Count")


pie(
  gender_count$Count,
  labels = paste0(
    gender_count$Gender, " (",
    round(gender_count$Count /
          sum(gender_count$Count) * 100),
    "%)"
  ),
  col = c("pink", "skyblue"),
  main = "Customer Gender Distribution"
)
```

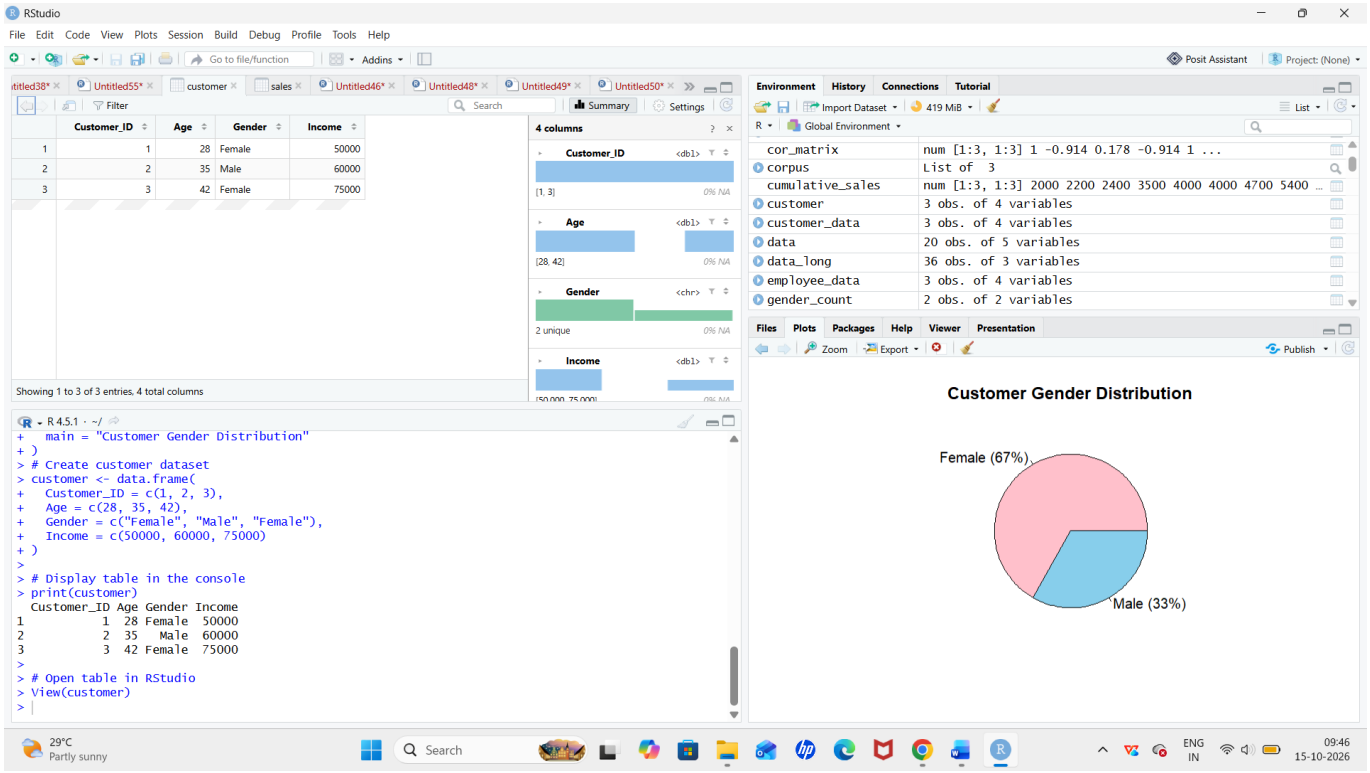


3. Build a table to show the demographic information for each customer. Label the table elements.

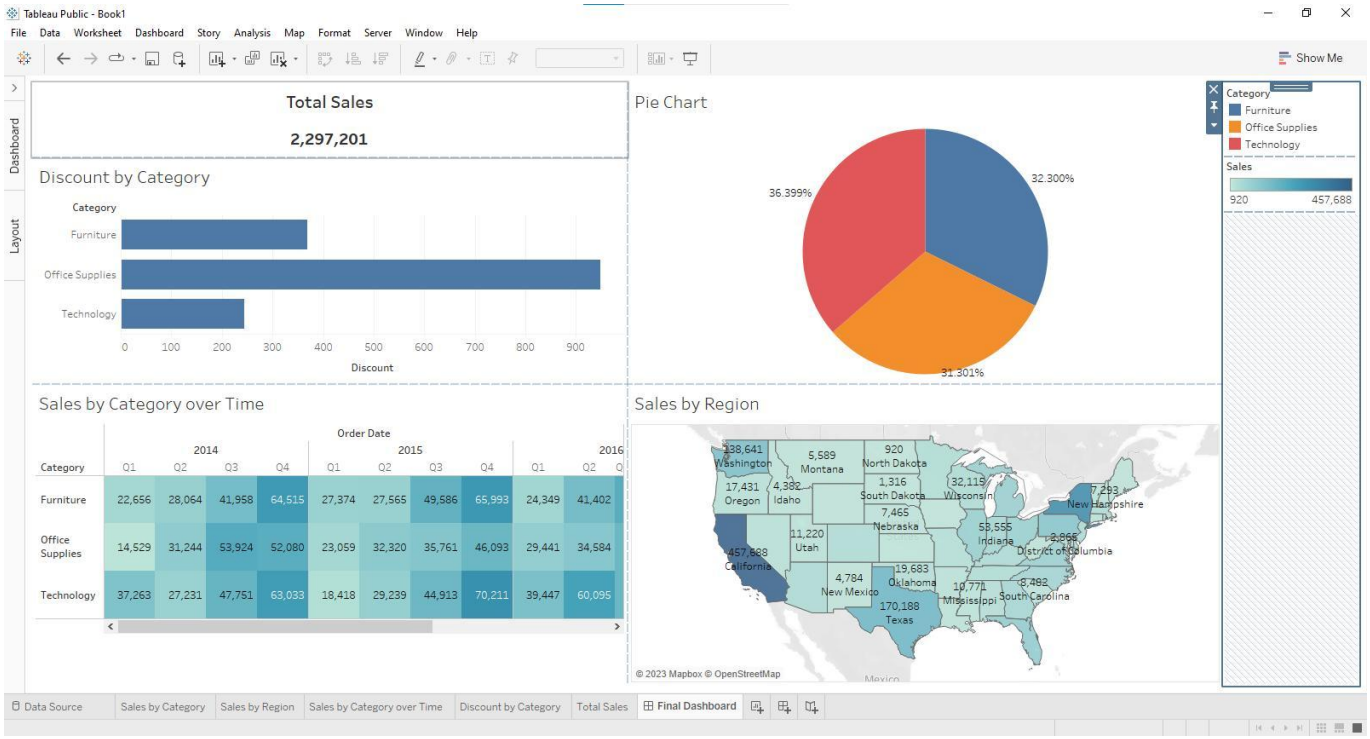
```
customer <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

print(customer)

View(customer)
```



4.Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive



PROCEDURE

1. Create a dataset containing Customer ID, Age, Gender, and Income.

2. Load the required R packages, such as ggplot2.

3. Create a bar chart to visualize customer ages, with appropriate axis labels and a chart title.

4. Generate a pie chart to represent the distribution of customers by gender.

5. Build a table displaying the demographic information of each customer.

6. Export the customer dataset as a CSV file using the write.csv() function.

7. Import the CSV file into Tableau and create separate worksheets for the bar chart, pie chart, and table.

8. Create a new Tableau dashboard and combine all three worksheets.

9. Add suitable titles, labels, and interactive filters to enable exploration of customer demographics.

Result

Thus, customer demographic analysis was performed using R programming by creating a bar chart for age distribution, a pie chart for gender distribution, and a table containing customer demographic details. The dataset was exported to CSV format and prepared for integration into an interactive Tableau dashboard, enabling users to explore customer ages, gender, and income effectively.